An explicit 6-regular graph of girth 9 on 4368 vertices

Claude (AI model by Anthropic)
run and submitted by Aleksandr Bersenev

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Abstract

We construct an explicit 6-regular simple graph of girth exactly 9 on 4368 vertices. It is a regular cover (voltage lift) of a two-vertex multigraph with voltages in $\mathrm{SL}(2,13)$. Consequently $n(6,9) \le 4368$, where $n(k,g)$ is the order of a (k,g)-cage. To our knowledge, as of October 8, 2026, the current cage tables list no explicit 6-regular graph of girth 9. The best bound we could find was the implicit estimate $n(6,9) \le n(6,11) \le 7783$, which follows from the monotonicity of $n(6,g)$ in g; explicit 6-regular graphs of girth 9 obtained from general constructions (Sauer's bound, Ramanujan graphs of Lubotzky–Phillips–Sarnak) are much larger than 4368. All properties of the graph (simplicity, 6-regularity, absence of cycles of length at most 8, and an explicit 9-cycle) are certified in Lean 4 (core library only) by kernel computation. The proof uses only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The lower bound $n(6,9) \ge 938$ is unchanged, so a large gap remains.

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1 Introduction

A (k,g)-*graph* is a k-regular graph of girth g. A (k,g)-*cage* is a (k,g)-graph of minimum order, and $n(k,g)$ denotes this order. For odd g, counting the vertices at distance at most $(g-1)/2$ from

a fixed vertex gives the Moore bound

$$n(k,g) \;\geq\; 1 + k \sum_{i=0}^{(g-3)/2} (k-1)^i.$$

For $(k,g) = (6,9)$ this is $1 + 6(1+5+25+125) = 937$; the current table lists the lower bound 938 [1, Table 2]. The cage problem asks for $n(k,g)$. Except for a few small parameter pairs it is wide open. Upper bounds come from explicit constructions, often found by computer. They are collected in the dynamic survey of Exoo and Jajcay [2] and, in a continuously updated form, in the cage meta-directory of the House of Graphs [3, 4].

The $(6,9)$ cell. In the summary table of upper bounds in the survey [2, Table 6], the row $k = 6$ has entries for $g \in \{5,6,7,8,12\}$ and none for $g \in \{9,10,11\}$. In 2026, Exoo, Goedgebeur, Jooken, Stubbe and Van den Eede [1] gave new upper bounds for eleven cases using exhaustive generation of lifts, tabu search, hill climbing and excision. Among them is the first nontrivial bound $n(6,11) \leq 7783$, obtained by excision from the $(6,12)$-cage on 7812 vertices (the incidence graph of a generalized hexagon of order 5). Their Table 2 lists the rows

$$(k,g) = (6,9):\ n \geq 938,\ n \leq ? \qquad \text{and} \qquad (k,g) = (6,10):\ n \geq 1563,\ n \leq ?,$$

and explains that "?" means a graph with these parameters is known to exist but is orders of magnitude larger than the lower bound. The cage table of the House of Graphs [4] (rows $k = 3,\dots,20$, columns $g = 5,\dots,20$, accessed October 8, 2026) also has empty cells at $(6,9)$ and $(6,10)$. It lists 312, 7783 and 7812 for $g = 8, 11, 12$.

The implicit bound. A theorem of Erdős and Sachs [5], stated in [2, Theorem 4], says that a k-regular graph of girth *at least* g with the minimum number of vertices has girth exactly g. Hence $n(k,g)$ is the minimum order of a k-regular graph of girth $\geq g$. Every (k,g')-graph with $g' \geq g$ has girth $\geq g$, so $n(k,g) \leq n(k,g')$. The survey [2, p. 5] also attributes this monotonicity to [6] and [7]. Combined with [1] it gives

$$n(6,9) \;\leq\; n(6,10) \;\leq\; n(6,11) \;\leq\; 7783. \tag{1.1}$$

(Sauer's strict inequality $n(k,g) < n(k,g+1)$ [2, Theorem 9] even gives $n(6,9) \leq 7781$.) This is the best upper bound on $n(6,9)$ that we found in the literature. It is not explicit: the argument does not exhibit a graph of girth 9.

Explicit graphs from general constructions. Explicit 6-regular graphs of girth 9 do exist in the literature, but only through general constructions whose outputs are far larger. Sauer's constructive proof gives $n(6,9) \leq 2 \cdot 4^7 = 32768$ [2, Theorem 7]. The Ramanujan graphs $X^{5,q}$ of Lubotzky, Phillips and Sarnak [10] are explicit 6-regular Cayley graphs of $\mathrm{PSL}(2,q)$ or $\mathrm{PGL}(2,q)$; a direct computation (breadth-first search) shows that $X^{5,13}$ and $X^{5,17}$, on 2184 and 4896 vertices, have girth 8, while $X^{5,29}$, on 12180 vertices, has girth exactly 9. None of these graphs appears in the cage tables, and all of them are larger than the graph constructed here.

Result. We give an explicit $(6,9)$-graph with $4368 = 2 \cdot |\mathrm{SL}(2,13)|$ vertices.

Theorem 1.1. *There is a 6-regular simple graph of girth exactly 9 on 4368 vertices. Consequently* $938 \leq n(6,9) \leq 4368$.

The graph is a voltage lift (regular cover) of a two-vertex base multigraph, with voltages in the group $\mathrm{SL}(2,13)$ of order 2184 (Section 3). Left multiplication by group elements acts by automorphisms. These automorphisms reduce the girth verification to the radius-4 balls around just two vertices. The voltages were found by a randomized backtracking search (Section 5). The full statement is formalized and proved in Lean 4 (Section 6).

To our knowledge, as of October 8, 2026, this is the first explicit $(6,9)$-graph recorded in the cage tables, and it is smaller than every explicit $(6,9)$-graph we are aware of (see the paragraph on general constructions above). It improves (1.1) by a factor of about 1.8. We make no claim of optimality. Our graph has 4.66 times as many vertices as the Moore bound, while the neighbouring records are 58, 270 and 1116 for $(3,9)$, $(4,9)$ and $(5,9)$ [1], at 1.26, 1.68 and 2.6 times their Moore bounds. Much smaller $(6,9)$-graphs very likely exist.

2 Definitions and statement

All graphs are finite, undirected and simple. A *cycle of length* $L \geq 3$ is a sequence $c_0, \ldots, c_{L-1}$ of pairwise distinct vertices with $c_i \sim c_{i+1}$ for $i < L-1$ and $c_{L-1} \sim c_0$. The *girth* is the minimum length of a cycle. A *non-backtracking walk* of length ℓ is a sequence $w_0, \ldots, w_\ell$ with $w_i \sim w_{i+1}$ and $w_{i+2} \neq w_i$.

Voltage lifts. Let B be a finite multigraph, possibly with loops. A *dart* is an edge with a chosen direction; every dart $d\colon u \to w$ has a reverse dart $d^{-1}\colon w \to u$. A loop gives two darts $u \to u$. A *voltage assignment* in a group Γ is a map α from darts to Γ with $\alpha(d^{-1}) = \alpha(d)^{-1}$. The *lift* B^α has vertex set $V(B) \times \Gamma$, and for each dart $d\colon u \to w$ and each $M \in \Gamma$ an edge $(u, M) \sim (w, M\alpha(d))$ (see [8]). A closed walk in B has a *net voltage*: the product of the voltages along it. A standard fact [8, 9] is that B^α has girth $\geq g$ iff every cyclically reduced closed walk of length $< g$ in B has net voltage $\neq 1$. We use this fact only to guide the search. The proof of Theorem 1.1 below is self-contained and does not depend on it.

3 The construction

Let $\Gamma = \mathrm{SL}(2,13)$, the group of 2×2 matrices over $\mathbb{F}_{13} = \mathbb{Z}/13\mathbb{Z}$ with determinant 1; $|\Gamma| = 13 \cdot (13^2 - 1) = 2184$. The base graph B has two vertices 0 and 1, one loop at each vertex, and four parallel edges between 0 and 1. Thus B is 6-regular, each loop counting twice. The voltages are

$$A = \left(\begin{smallmatrix} 0 & 9 \\ 10 & 2 \end{smallmatrix}\right) \text{ (loop at 0)}, \qquad C = \left(\begin{smallmatrix} 1 & 0 \\ 8 & 1 \end{smallmatrix}\right) \text{ (loop at 1)},$$

$$E_1 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right), \quad E_2 = \left(\begin{smallmatrix} 1 & 4 \\ 4 & 4 \end{smallmatrix}\right), \quad E_3 = \left(\begin{smallmatrix} 9 & 11 \\ 1 & 10 \end{smallmatrix}\right), \quad E_4 = \left(\begin{smallmatrix} 5 & 4 \\ 9 & 10 \end{smallmatrix}\right) \qquad \text{(edges } 0 \to 1\text{)}.$$

All six matrices have determinant 1 modulo 13. Table 1 lists the twelve darts with their voltages, including the reverse darts with the inverse voltages. This is exactly the table `darts` of the Lean file.

Let $X = B^\alpha$. Concretely,

$$V(X) = \{0,1\} \times \mathrm{SL}(2,13), \qquad |V(X)| = 4368,$$

and the six neighbours of a vertex are

$$N(0,M) = \{(0,MA),\ (0,MA^{-1}),\ (1,ME_1),\ (1,ME_2),\ (1,ME_3),\ (1,ME_4)\},$$
$$N(1,M) = \{(1,MC),\ (1,MC^{-1}),\ (0,ME_1^{-1}),\ (0,ME_2^{-1}),\ (0,ME_3^{-1}),\ (0,ME_4^{-1})\}.$$

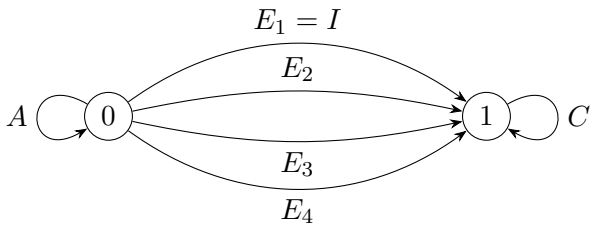


Figure 1: The base multigraph B with its voltages in $\mathrm{SL}(2,13)$.

dart at 0	head	voltage	dart at 1	head	voltage
A	0	$\left(\begin{smallmatrix} 0 & 9 \\ 10 & 2 \end{smallmatrix}\right)$	C	1	$\left(\begin{smallmatrix} 1 & 0 \\ 8 & 1 \end{smallmatrix}\right)$
A^{-1}	0	$\left(\begin{smallmatrix} 2 & 4 \\ 3 & 0 \end{smallmatrix}\right)$	C^{-1}	1	$\left(\begin{smallmatrix} 1 & 0 \\ 5 & 1 \end{smallmatrix}\right)$
E_1	1	$\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$	E_1^{-1}	0	$\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$
E_2	1	$\left(\begin{smallmatrix} 1 & 4 \\ 4 & 4 \end{smallmatrix}\right)$	E_2^{-1}	0	$\left(\begin{smallmatrix} 4 & 9 \\ 9 & 1 \end{smallmatrix}\right)$
E_3	1	$\left(\begin{smallmatrix} 9 & 11 \\ 1 & 10 \end{smallmatrix}\right)$	E_3^{-1}	0	$\left(\begin{smallmatrix} 10 & 2 \\ 12 & 9 \end{smallmatrix}\right)$
E_4	1	$\left(\begin{smallmatrix} 5 & 4 \\ 9 & 10 \end{smallmatrix}\right)$	E_4^{-1}	0	$\left(\begin{smallmatrix} 10 & 9 \\ 4 & 5 \end{smallmatrix}\right)$

Table 1: The twelve darts of B and their voltages (entries modulo 13).

Some orders, for orientation: A and C have order 13, so the loops lift to 13-cycles. The elements $E_iE_j^{-1}$ $(i<j)$ have orders $14, 14, 13, 26, 12, 26$, so the digons of B lift to cycles of length between 24 and 52. These numbers are not used in the proof.

4 Proof of Theorem 1.1

Lemma 4.1 (simple and 6-regular)**.** *X is a simple 6-regular graph.*

Proof. If (w, MX) is a neighbour of (u, M) via a dart with voltage X, then $(u, M) = (u, (MX)X^{-1})$ is a neighbour of (w, MX) via the reverse dart. So adjacency is symmetric. For fixed u the six voltages of the darts at u are pairwise distinct (Table 1), so the six neighbours (w, MX) are distinct by left cancellation. None equals (u, M), since no dart from u to u has voltage I. Hence every vertex has exactly six distinct neighbours, all different from itself. □

Lemma 4.2 (automorphisms)**.** *For every $G \in \mathrm{SL}(2,13)$ the map $\lambda_G\colon (u, M) \mapsto (u, GM)$ is an automorphism of X. The maps λ_G act transitively on each fibre $\{u\} \times \mathrm{SL}(2,13)$.*

Proof. By associativity $\lambda_G(w, MX) = (w, G(MX)) = (w, (GM)X)$, so λ_G maps the dart-d neighbour of (u, M) to the dart-d neighbour of $\lambda_G(u, M)$. Its inverse is $\lambda_{G^{-1}}$. Moreover $\lambda_M(u, I) = (u, M)$. □

Lemma 4.3 (local girth criterion)**.** *Let r be a vertex of a graph and $R \geq 1$. Suppose the non-backtracking walks of length at most R starting at r have pairwise distinct end vertices. Then no cycle of length $L \leq 2R$ passes through r.*

Proof. Suppose $c_0 = r, c_1, \ldots, c_{L-1}$ is a cycle with $3 \leq L \leq 2R$, and let $a = \lfloor L/2 \rfloor$. The walks $W_1 = (c_0, c_1, \ldots, c_a)$ and $W_2 = (c_0, c_{L-1}, c_{L-2}, \ldots, c_a)$ have lengths $a \leq R$ and $L - a \leq R$. They are non-backtracking because the c_i are distinct and $L \geq 3$. Both end at c_a. They are different walks: if $a \neq L - a$ their lengths differ, and otherwise their second vertices $c_1 \neq c_{L-1}$ differ. This contradicts the hypothesis. □

Proposition 4.4 (the two root balls)**.** *For $r \in \{(0,I),(1,I)\}$, the $1+6+30+150+750 = 937$ non-backtracking walks of length at most 4 in X starting at r have pairwise distinct end vertices.*

Proof. This is a finite computation. It enumerates 2×937 walks, each step being a 2×2 matrix product modulo 13. We certified it with the Lean kernel (theorems `root_0`, `root_1` in Section 6). It was also checked independently in Python by breadth-first search from *every* vertex of X (Section 5). □

Proposition 4.5 (a 9-cycle)**.** *The following nine vertices form a cycle in X:*

$$(0,I),\ (0,\left(\begin{smallmatrix}0&9\\10&2\end{smallmatrix}\right)),\ (0,\left(\begin{smallmatrix}12&5\\7&3\end{smallmatrix}\right)),\ (0,\left(\begin{smallmatrix}11&1\\4&4\end{smallmatrix}\right)),\ (1,\left(\begin{smallmatrix}9&1\\1&6\end{smallmatrix}\right)),$$
$$(1,\left(\begin{smallmatrix}1&1\\5&6\end{smallmatrix}\right)),\ (0,\left(\begin{smallmatrix}1&1\\9&10\end{smallmatrix}\right)),\ (0,\left(\begin{smallmatrix}5&4\\9&10\end{smallmatrix}\right)),\ (1,\left(\begin{smallmatrix}5&4\\9&10\end{smallmatrix}\right)).$$

Proof. The consecutive steps, including the closing step back to $(0,I)$, use the darts

$$A,\ A,\ A,\ E_3,\ C^{-1},\ E_4^{-1},\ A^{-1},\ E_1,\ E_4^{-1}$$

in this order. One checks each matrix product directly; for instance $A^2 = \left(\begin{smallmatrix}12&5\\7&3\end{smallmatrix}\right)$ and $A^3 = \left(\begin{smallmatrix}11&1\\4&4\end{smallmatrix}\right)$. The nine vertices are visibly distinct. □

Proof of Theorem 1.1. By Theorem 4.1, X is simple and 6-regular on 4368 vertices. Let c be any cycle of length $L \le 8$, and let (u,M) be one of its vertices. By Theorem 4.2, $\lambda_{M^{-1}}$ maps c to a cycle of the same length through (u,I). This contradicts Theorem 4.3 with $R = 4$ and Theorem 4.4. Hence the girth is at least 9. By Theorem 4.5 it is exactly 9. The lower bound 938 is quoted from [1]. □

Remark 4.6 (connectivity and diameter)*.* A breadth-first search shows that X is connected with diameter 7. Connectivity is not needed for the cage problem, and neither fact is formalized.

Remark 4.7 (a $(6,10)$-graph on 8736 vertices)*.* By Sachs' construction for odd g [2, Theorem 5], the canonical double cover $X \times K_2$ is a 6-regular bipartite graph of girth 10 on 8736 vertices; we confirmed the girth by computer. This does not improve anything, because $8736 > 7783 \ge n(6,10)$ by (1.1). We mention it only because it is explicit.

5 How the voltages were found

The criterion of Section 2 turns the search into a constraint satisfaction problem. Up to rotation and reversal, the base graph B has 31,734 cyclically reduced closed walks of length at most 8, and each one must receive a non-identity net voltage. After normalizing one parallel edge to $E_1 = I$, five voltages remain free. We assigned them one at a time by a randomized depth-first search. When a variable is reached, every closed walk that contains it exactly once forbids exactly one value, which is computed directly. Walks that contain it several times are tested against each surviving candidate. This is the same principle as the lift-generation algorithms of [1], run as a randomized rather than exhaustive search.

We chose $\mathrm{SL}(2,q)$ for two reasons. It has few elements of small order (a single involution), and it makes the formal verification simple (Theorem 4.2). For $\mathrm{SL}(2,13)$ a solution appeared within seconds (about $5 \cdot 10^3$ search nodes). The solution was then checked by an independent Python program that builds X explicitly and runs breadth-first search from all 4368 vertices; it reports 6-regular, simple, symmetric, girth 9. A second, independently written script confirmed the same.

6 Formal verification

The file `lean/Main.lean` (about 730 lines) is self-contained. It uses Lean 4.34.1 [11] (x86-64 Linux release) with the core library and `Std` only, and no Mathlib. The definitions used in the main statement, verbatim:

```
/-- A graph on `Fin n` given by a Boolean adjacency relation is *simple* if the relation is
symmetric and irreflexive. -/
def IsSimpleGraph {n : Nat} (adj : Fin n → Fin n → Bool) : Prop :=
  (∀ u v, adj u v = adj v u) ∧ (∀ v, adj v v = false)

/-- degree of `v`: the number of vertices `w` with `adj v w`. -/
def degree {n : Nat} (adj : Fin n → Fin n → Bool) (v : Fin n) : Nat :=
  ((List.finRange n).filter (fun w => adj v w)).length

/-- `k`-regular: every vertex has degree `k`. -/
def IsRegular {n : Nat} (adj : Fin n → Fin n → Bool) (k : Nat) : Prop :=
  ∀ v, degree adj v = k

/-- `c = [c_0, ..., c_{L-1}]` is a cycle: at least 3 pairwise distinct vertices with
`c_i ~ c_{(i+1) mod L}` for all `i`. -/
def IsCycle {α : Type} (adj : α → α → Bool) (c : List α) : Prop :=
  3 ≤ c.length ∧ c.Nodup ∧
    ∀ i (h : i < c.length), adj c[i] (c[(i + 1) % c.length]'(Nat.mod_lt _ (Nat.lt_of_le_of_lt
    ↪  (Nat.zero_le _) h))) = true


/-- girth exactly `g`: some cycle has length `g`, and every cycle has length at least `g`. -/
def HasGirth {n : Nat} (adj : Fin n → Fin n → Bool) (g : Nat) : Prop :=
  (∃ c, IsCycle adj c ∧ c.length = g) ∧ ∀ c, IsCycle adj c → g ≤ c.length
```

The main theorem, verbatim:

```
theorem main : ∃ adj : Fin 4368 → Fin 4368 → Bool,
    IsSimpleGraph adj ∧ IsRegular adj 6 ∧ HasGirth adj 9 :=
  ⟨adjF, simple, regular, ⟨exists_cycle, no_short_cycles⟩⟩
```

Proof architecture.

1. *Generic part.* This part does not depend on the data:
   - a bitset duplicate test `allDistinct` with a soundness proof;
   - an enumeration `layer` of reversed non-backtracking walks, with a completeness proof;
   - the theorem `no_short_cycle`, the formal version of Theorem 4.3: if the end vertices of all enumerated walks of length $\leq R$ from r are distinct, no cycle of length $\leq 2R$ starts at r.
2. *Matrices modulo p.* The reduced product `mmul` is associative for arbitrary natural-number entries (`mmul_assoc`; the polynomial identity is closed by `grind`). The identity laws hold for reduced entries.
3. *The lift.* The definitions are:

- vertices are pairs (u, M);
- `nbrs` is given by the dart table of Table 1;
- `verts` is the list of all (u, M) with $u < 2$, entries < 13 and $\det M \equiv 1$, obtained by filtering all $2 \cdot 13^4$ tuples.

Symmetry of adjacency is proved algebraically from a kernel check of the dart table (every dart has a reverse dart with inverse voltage). The automorphisms are $\lambda_G(u, M) = (u, GM)$, and injectivity uses $MM^{-1} = I$.

4. *Finite checks, all by* `decide +kernel`.

   - The dart table.
   - 26 block checks `cc_0`, ..., `cc_25`. They cover the vertex count 4368; closure of $\{0, 1\} \times \mathrm{SL}(2, 13)$ under the six neighbour maps; distinctness of each vertex and its six neighbours; and $M^{-1}M = MM^{-1} = I$ for every vertex.
   - The two root-ball checks `root_0` and `root_1` (Theorem 4.4).
   - The explicit 9-cycle of Theorem 4.5.

5. *Transfer to* `Fin 4368`. The vertex list has no duplicates and has length 4368, so indexing gives a bijection with `Fin 4368`. The adjacency `adjF` is pulled back along it. Degrees are computed through a permutation argument (`List.perm_ext_iff_of_nodup`).

Statement audit.

- The vertex type is `Fin 4368`, so the graph has exactly 4368 vertices.
- `IsSimpleGraph` is symmetry plus irreflexivity of a Boolean relation, so there are no loops and no multiple edges.
- `degree` counts the w in `List.finRange n`, which lists each vertex exactly once, with `adj v w = true`. This is the usual degree.
- `IsCycle` requires at least 3 pairwise distinct vertices with consecutive adjacency, including the closing pair $c_{L-1} \sim c_0$ (index $(i + 1) \bmod L$). These are exactly the cycles of a simple graph.
- `HasGirth adj 9` asserts both that a cycle of length 9 exists and that every cycle has length ≥ 9. This is "girth exactly 9"; it is not vacuous, because a 9-cycle must exist.
- The theorem has no hypotheses, and the existential quantifier is outermost.

Axioms and resources. The file ends with `#print axioms Cage.main`; the output, verbatim, is

```
'Cage.main' depends on axioms: [propext, Classical.choice, Quot.sound]
```

There is no `sorry`, no `axiom` declaration and no `native_decide`. All finite computations are checked by the kernel. On a shared machine (load average about 15 on 14 cores), the whole file checks in 153.6 s wall time with peak memory 0.99 GB. The file sets `set_option Elab.async false`. With parallel elaboration, the concurrent kernel checks failed to create threads under the memory limits of our checking wrapper. The option only changes scheduling, not what is checked.

7 Limitations and open questions

- *Novelty.* The claim that no explicit $(6, 9)$-graph was recorded in the cage tables rests on the sources [2, 1, 4] and on literature searches, all as of October 8, 2026. We cannot rule out unpublished or unindexed constructions. Explicit $(6, 9)$-graphs from general constructions (Sauer's bound, Ramanujan graphs) are known but much larger; see Section 1.
- *Quality of the bound.* 4368 is 4.66 times the Moore bound. It is the first explicit value, not a strong one; the gap $938 \leq n(6, 9) \leq 4368$ is wide.
- *Smaller lifts we could not find.* Several searches failed:
  - the same two-vertex base over $\mathrm{SL}(2, 11)$, which would give 2640 vertices: about 40 minutes of randomized backtracking and 10 minutes of min-conflicts local search; the best assignment had 5 violated closed walks;
  - random Cayley graphs (three generators and their inverses) of $\mathrm{PSL}(2, 17)$, $\mathrm{PSL}(2, 19)$, $\mathrm{PGL}(2, 13)$ and $\mathrm{SL}(2, 13)$: none of girth 9 in 1.5–10 million trials each;
  - random Cayley graphs of $\mathrm{SL}(2, 17)$: girth 9 occurs (about $2 \cdot 10^{-4}$ of trials), but these have $4896 > 4368$ vertices.

  Whether $\mathrm{SL}(2, 11)$-lifts or other small lifts of order about 2600–3300 exist remains open.
- *The* $(6, 10)$ *cell.* We also tried to construct a $(6, 10)$-graph of order below the implicit bound 7783. None of the following produced a solution:
  - bipartite lifts of the dipole (two vertices joined by six parallel edges) over $\mathrm{SL}(2, 13)$, which would give 4368 vertices;
  - the same dipole over the group of matrices of determinant ± 1 in $\mathrm{GL}(2, 11)$, which would give 5280 vertices;
  - lifts of 4-vertex bipartite bases over $\mathrm{SL}(2, 11)$, which would give 5280 vertices;
  - canonical double covers of $\mathrm{SL}(2, 11)$-lifts of the base B, which would also give 5280 vertices; here local search with exhaustive two-variable repair stalled at exactly one violated closed walk in four independent runs;
  - random bipartite double covers of Cayley graphs of $\mathrm{SL}(2, 13)$: $5 \cdot 10^7$ trials, girth at most 8 in all of them.

  So the $(6, 10)$ cell still has no explicit graph below 7783. A Lean template for the bipartite girth-10 case, with a formal lemma that every cycle in a properly 2-coloured graph has even length, is prepared but has no certified data.

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